



Literature Review Document

Course Title: MSc Data Science and AI

Project Title: An End-to-End Data Engineering and Predictive Framework for Demand Forecasting and Inventory Optimization in Pharmaceutical Distribution

Student Name: Shiyas Rasi

Student ID: M01087756

1. Introduction

Pharmaceutical distribution operates under conditions that make demand forecasting and inventory control unusually difficult. Demand can be seasonal, intermittent, geographically uneven, and sensitive to epidemiological events, while inventory decisions are constrained by product shelf life, service-level expectations, lead-time uncertainty, and the financial cost of holding excess stock. In this context, inaccurate forecasts can contribute to both stockouts and avoidable expiry-related waste. The problem is therefore not only to improve predictive accuracy, but also to translate forecasts into reliable replenishment decisions.

The forecasting literature has evolved from classical statistical methods, including exponential smoothing and ARIMA-family models, towards machine-learning and deep-learning approaches that can represent nonlinear relationships and use a wider set of explanatory variables. Research on pharmaceutical demand has compared multiple statistical and machine-learning approaches, while the broader intermittent-demand literature shows that sparse and irregular demand requires careful model selection, evaluation, and inventory-aware validation (Nikolopoulos et al., 2016; Türkmen et al., 2021; Li et al., 2022; Yani et al., 2023).

However, predictive performance alone does not establish operational value. A forecasting model depends on the quality, consistency, and timeliness of upstream data, while its outputs must be integrated with downstream inventory rules such as safety stock and reorder points. This review therefore examines four connected themes: pharmaceutical demand characteristics; classical forecasting methods; machine-learning and deep-learning approaches; and the operational integration of forecasting, data engineering, and inventory optimization. The purpose is to establish the conceptual and methodological basis for the proposed Pharma-Flow framework and to identify a defensible research gap.

2. Literature Review Methodology

A structured literature review approach was used to identify research relevant to pharmaceutical demand forecasting, intermittent demand, machine learning, inventory optimization, and data-engineering operationalization. Searches were designed around combinations of terms such as “pharmaceutical demand forecasting”, “drug demand prediction”, “intermittent demand forecasting”, “ARIMA”, “SARIMA”, “XGBoost”, “LSTM”, “inventory optimization”, “safety stock”, “reorder point”, “ETL pipeline”, and “MLOps”. Sources were considered from academic databases and indexing services including Google Scholar, IEEE Xplore, ScienceDirect, and publisher databases.

The inclusion criteria prioritized peer-reviewed journal articles and conference papers that: (1) addressed demand forecasting, inventory planning, or supply-chain prediction; (2) reported a quantitative modelling method; (3) provided a relevant comparison of statistical, machine-learning, or deep-learning techniques; or (4) examined intermittent demand or the connection between forecasting and inventory decisions. Technical documentation was considered only where it contributed directly to architectural or implementation considerations. Purely conceptual supply-chain discussions without a forecasting, data, or inventory-control component were excluded.

For each selected study, the review considered the application domain, data characteristics, forecasting horizon, model family, feature design, evaluation metrics, and relationship between forecast outputs and inventory decisions. The synthesis is thematic rather than a simple paper-by-paper summary. This is important because the proposed project combines several disciplines—data

engineering, forecasting, and inventory optimization—and the research gap lies partly in the integration between these areas.

3. Evolution of Demand Forecasting Research

3.1 Classical statistical forecasting

Classical forecasting methods remain important because they provide transparent and computationally efficient baselines. Naïve forecasts, moving averages, exponential smoothing, ARIMA, and seasonal ARIMA models are widely used where historical demand exhibits stable level, trend, and seasonality. In pharmaceutical forecasting, Nikolopoulos et al. (2016) compared a range of methods for branded and generic pharmaceutical series, including ARIMA, exponential smoothing, naïve methods, regression, and diffusion models. This work illustrates an important point: no single forecasting family is universally optimal, and pharmaceutical series can differ substantially according to product lifecycle and market behaviour.

The principal strength of ARIMA-family models is their explicit treatment of autocorrelation and, in seasonal variants, recurring temporal structure. They are interpretable and suitable as benchmarks. Their limitations become more visible when demand is highly sparse, lumpy, or influenced by nonlinear interactions among multiple variables. In such cases, aggregation level and forecast horizon can materially change performance. Consequently, the proposed project should retain SARIMA as a baseline rather than assume in advance that a machine-learning model will always be superior.

3.2 Intermittent and lumpy demand

Intermittent demand is especially relevant to pharmaceutical distribution because some SKUs may have many zero-demand periods followed by irregular positive orders. Standard accuracy measures and conventional forecasting models can behave poorly in this setting. Türkmen et al. (2021) proposed a renewal-process perspective that generalizes Croston-type approaches and supports more flexible modelling of demand arrivals. Rožanec and Mladenović (2021) similarly reframed lumpy-demand forecasting as two related tasks: predicting whether demand will occur and estimating demand size.

Li et al. (2022) examined feature-based forecast combinations for intermittent demand and emphasized not only point accuracy but also bias, quantile forecasting, computational efficiency, and inventory implications. These findings are relevant to PharmaFlow because a single global regression model trained across all SKUs may conceal important differences between fast-moving and intermittent products. A stronger design is therefore to classify demand patterns or product segments first, then compare forecasting performance by segment and evaluate whether grouping low-frequency products improves or weakens operational results.

3.3 Machine learning and gradient boosting

Machine-learning methods can incorporate calendar effects, lagged demand, rolling statistics, product attributes, customer or region information, and other exogenous variables. Tree-based ensemble methods are particularly attractive for tabular transactional data because they can model nonlinear relationships and interactions without requiring the same distributional assumptions as classical time-series models. XGBoost is a regularized gradient-boosting framework designed for efficient and scalable tree learning (Chen and Guestrin, 2016).

In pharmaceutical demand research, recent comparative studies increasingly evaluate machine-learning approaches alongside statistical baselines. Yani et al. (2023) examined demand-forecasting accuracy in the pharmaceutical supply chain and highlighted the relevance of machine-learning algorithms to this domain. Ascensão (2023) compared XGBoost, LSTM, and Prophet using pharmaceutical sales data and a structured CRISP-DM process. Such studies support the inclusion of XGBoost in the proposed framework, but they do not justify treating it as automatically superior. Performance depends on data granularity, feature engineering, horizon, product heterogeneity, and validation design.

For PharmaFlow, XGBoost is methodologically suitable because the source data are transactional and can be transformed into lagged and rolling features. Nevertheless, leakage prevention is essential. Random train-test splitting would be inappropriate for time-dependent demand data because future observations could indirectly influence model training. The project should use chronological holdout testing or rolling-origin evaluation, with all lag and rolling-window features computed using only information available before each forecast origin.

3.4 Deep learning approaches

LSTM networks and related deep-learning architectures are designed to learn temporal dependencies and have been applied to demand forecasting. Their potential advantage is the ability to represent complex sequential relationships, particularly when large and sufficiently regular datasets are available. However, deep-learning models generally introduce greater computational cost, tuning complexity, and lower interpretability than many classical or tree-based alternatives.

The literature does not support a universal claim that LSTM is either always better or always worse than gradient boosting. Comparative performance is context-dependent. For a medium-scale MSc project based on five years of B2B transaction data, the most defensible approach is to use a strong statistical baseline and one or more well-justified machine-learning models, while treating deep learning as an optional extension if data volume and time permit. This keeps the evaluation aligned with the research objectives rather than expanding the model set without clear analytical value.

4. From Forecast Accuracy to Inventory Decisions

A major theme emerging from the literature is that forecasting and inventory control should not be evaluated independently. A model with a lower average forecast error does not necessarily produce the lowest stockout rate or holding cost. Forecast bias, lead-time uncertainty, service-level targets, and demand variability all influence inventory performance. Research on intermittent demand also demonstrates the value of examining inventory implications rather than relying exclusively on aggregate error measures (Li et al., 2022).

The proposed project therefore has a strong opportunity to evaluate two layers of performance. The first is predictive performance, using metrics such as MAE, RMSE, and a percentage-based measure selected carefully for zero-demand series. The second is operational performance, using historical back-testing to compare stockout frequency, service level or fill rate, average inventory, excess stock, and expiry-risk proxies where expiry information is available. This dual evaluation would make the contribution stronger than a study that reports forecasting error alone.

The inventory layer should also define its assumptions explicitly. A reorder point can be represented conceptually as expected demand during lead time plus safety stock. The exact safety-stock formulation depends on whether uncertainty is attributed to demand, lead time, or both.

PharmaFlow should therefore document the chosen service level, lead-time assumptions, demand variability window, and replenishment review policy. Without these assumptions, the connection between prediction and replenishment would remain under-specified.

5. Data Engineering and Operationalization

Forecasting research often begins with a modelling-ready dataset, whereas operational systems must create that dataset repeatedly and consistently. In a real distribution ledger, likely engineering challenges include duplicate transactions, inconsistent product identifiers, missing timestamps, returns or cancellations, unit-of-measure inconsistencies, changes in product master data, and irregular transaction timing. These issues directly affect feature quality and model validity.

The proposed framework addresses this problem by designing a reproducible pipeline from raw transactional data to analytical features and forecasts. A relational database such as PostgreSQL can support staging, cleaned, and analytical layers. The pipeline should include data-quality checks, documented transformation rules, reproducible feature generation, model-version metadata, and a clear separation between training and inference datasets. These components are more academically meaningful when they are evaluated rather than merely implemented.

Accordingly, the data-engineering contribution should be assessed using measurable criteria such as pipeline reproducibility, data-quality rule coverage, processing time, failure handling, and the ability to rerun the workflow on new data without manual restructuring. This framing strengthens the claimed operationalization gap: the contribution is not simply the use of PostgreSQL or ETL, but the empirical integration and evaluation of a repeatable data-to-decision workflow.

6. Critical Synthesis and Research Gap

The reviewed literature provides substantial evidence for three separate propositions. First, pharmaceutical and related supply-chain demand can exhibit heterogeneous patterns that require more than one forecasting strategy. Second, machine-learning methods can exploit nonlinear relationships and engineered features, but their advantage over statistical methods is data- and context-dependent. Third, intermittent-demand research shows that model evaluation should consider demand occurrence, bias, forecast uncertainty, and inventory consequences.

The gap addressed by this project should therefore be stated carefully. It would be too broad to claim that existing research systematically ignores data pipelines or inventory integration. A more defensible gap is that comparatively fewer empirical studies evaluate, within one reproducible pharmaceutical-distribution workflow, the full chain from raw multi-SKU transactional data, through governed preprocessing and feature engineering, to time-aware model comparison and inventory-policy back-testing.

PharmaFlow addresses this integration gap through four linked contributions: a reproducible ETL/ELT pipeline for five years of multi-SKU B2B transaction data; a time-aware comparison of SARIMA and gradient-boosting models; segment-aware analysis of heterogeneous and intermittent demand; and an inventory simulation layer that converts forecasts into replenishment decisions and evaluates operational outcomes. The novelty is therefore located in integration, reproducibility, and operational evaluation rather than in proposing a new forecasting algorithm.

7. Implications for the Proposed Methodology

The literature review leads to several concrete methodological decisions. The project should first profile demand at SKU and aggregate levels, including zero-demand frequency, intermittency, seasonality, trend, and volume concentration. Product segmentation should be considered before model training because high-volume and intermittent items may require different treatment.

Second, model evaluation should use time-series cross-validation or rolling-origin evaluation. SARIMA should serve as an interpretable baseline, while XGBoost or CatBoost can be evaluated using lagged demand, rolling statistics, calendar variables, and relevant product or client features. Hyperparameter tuning must be nested within the temporal validation design to avoid optimistic estimates.

Third, evaluation metrics should be chosen with the data characteristics in mind. MAE and RMSE are useful scale-dependent measures, but MAPE can be unstable or undefined when actual demand is zero. Where intermittent demand is substantial, the study should consider scaled or zero-safe measures and report results by demand segment rather than only as a global average.

Finally, inventory back-testing should compare a baseline replenishment policy with forecast-driven policies under documented lead-time and service-level assumptions. The final dashboard should communicate both forecast quality and operational outcomes, thereby supporting decision-making rather than functioning only as a visualization of predicted values.

8. Conclusion

The literature indicates a clear progression from classical statistical forecasting towards machine-learning and deep-learning approaches capable of representing nonlinear demand patterns and incorporating richer features. Nevertheless, the evidence also shows that model superiority is conditional: data characteristics, intermittency, forecast horizon, feature design, and validation strategy all affect results. For pharmaceutical distribution, forecasting should therefore be treated as one component of a broader inventory decision system.

The proposed Pharma-Flow project is well positioned to contribute through an integrated and reproducible framework that links raw transactional data, data engineering, demand segmentation, time-aware model comparison, and inventory-policy evaluation. The strongest research contribution is not a claim that one algorithm universally outperforms another, but an empirical demonstration of how forecasting methods can be operationalized and assessed end to end on real multi-SKU pharmaceutical distribution data. This revised framing aligns the literature review more closely with the project proposal and provides a clearer basis for the methods, evaluation plan, and intended contribution.

9. References

- Ascensão, J. A. L. S. C. (2023). Forecasting Demand in the Pharmaceutical Industry Using Machine Learning Models. NOVA Information Management School.
- Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794.
- Li, L., Kang, Y., Petropoulos, F. and Li, F. (2022). Feature-based intermittent demand forecast combinations: bias, accuracy and inventory implications. arXiv:2204.08283.
- Nikolopoulos, K., Syntetos, A. A., Boylan, J. E., Petropoulos, F. and Assimakopoulos, V. (2016). Forecasting branded and generic pharmaceuticals. International Journal of Forecasting, 32(2), 344–357.
- Rožanec, J. M. and Mladenčić, D. (2021). Reframing demand forecasting: a two-fold approach for lumpy and intermittent demand. arXiv:2103.13812.
- Türkmen, A. C., Januschowski, T., Wang, Y. and Cemgil, A. T. (2021). Intermittent demand forecasting with renewal processes. International Journal of Forecasting, 37(2), 742–755.
- Yani, L. P. E. et al. (2023). Demand forecasting accuracy in the pharmaceutical supply chain: a machine learning approach. International Journal of Pharmaceutical and Healthcare Marketing, 17(1).